Azmari Sultana

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$y \rightarrow$ actual

$\hat{y} \rightarrow$ prediction

$$P(y=1|x) = \sigma(z) = \hat{y}$$

$$P(y=0|x) = 1-\sigma(z) = 1-\hat{y}$$

$$\boxed{P(y|x) = (\hat{y})^{y}(1-\hat{y})^{1-y}}$$ $\rightarrow$ for, 1 data

$y=1 \Rightarrow$

$$P(y=1|x) = (\hat{y})^{1} \times (1-\hat{y})^{1-1} = \hat{y}$$

$y=0 \Rightarrow$

$$P(y=0|x) = (\hat{y})^{0} \times (1-\hat{y})^{1-0} = (1-\hat{y})$$

Total probability / Likelihood:

$$P(y^{(1)}, y^{(2)}, y^{(3)}, \dots \dots y^{(m)} | x^{(1)}, x^{(2)}, x^{(3)}, \dots x^{(m)})$$

$$= P(y^{(1)}|x^{(1)})\, P(y^{(2)}|x^{(2)})\, P(y^{(3)}|x^{(3)}) \dots \dots P(y^{(m)}|x^{(m)})$$

$$= \prod_{i=1}^{m} P(y^{(i)}|x^{(i)})$$

$$= \prod_{i=1}^{m} (\hat{y}^{(i)})^{y^{(i)}} \times (1-\hat{y}^{(i)})^{1-y^{(i)}}$$ $\rightarrow$ likelihood

$$\text{log likelihood} = \sum_{i=1}^{m} \log\left[(\hat{y}^{(i)})^{y^{(i)}} \times (1-\hat{y}^{(i)})^{1-y^{(i)}}\right]$$

$\downarrow$

calc এ ln use করবো

$$= \sum_{i=1}^{m} \log(\hat{y}^{(i)})^{y^{(i)}} + \log(1-\hat{y}^{(i)})^{1-y^{(i)}}$$

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$$= \sum_{i=1}^{m} y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})$$

(-) minus দিয়ে দিবো

minimize করার goal থাকবে → loss function

Cross Entropy

$$L_{CE} = - \sum_{i=1}^{m} y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})$$

x	y	$\hat{y}$	
$x^{(1)}$	1	1	→ Loss = 0
$x^{(2)}$	1	0	→ missclassified
$x^{(3)}$	0	0	→ Loss = 0
$x^{(4)}$	0	1	→ missclassified

$x^{(1)}$: $y = 1$, $\hat{y} = 1$

$$L_{CE} = - y^{(i)} \log(\hat{y}^{(i)}) - (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})$$

$$= -1 \log(1) - (1 - 1) \log(1 - 1)$$

$$= -\log 1 = 0$$

correctly classified

So, Loss = 0

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$x^{(2)}$: $y = 1$, $\hat{y} = 0$

$$L_{CE} = -1\log(0) - (1-1)\log(1-0)$$
$$= \infty$$

↳ high loss

Derivative of loss function

$$L_{CE} = -y\log\hat{y} - (1-y)\log(1-\hat{y})$$

$$z = w_1x_1 + w_2x_2 + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}}$$

$$w_1 = w_1 - \eta\frac{\delta L_{CE}}{\delta w_1}$$

$$w_2 = w_2 - \eta\frac{\delta L_{CE}}{w_2}$$

$$b = b - \eta\frac{\delta L_{CE}}{\delta b}$$

$$\frac{\delta L_{CE}}{\delta b} = \frac{\delta L_{CE}}{\delta\hat{y}} \times \frac{\delta\hat{y}}{\delta z} \times \frac{\delta z}{\delta b}$$

$$\frac{\delta L_{CE}}{\delta\hat{y}} = -\frac{\delta}{\delta\hat{y}}(y\log\hat{y}) - \frac{\delta}{\delta\hat{y}}\left[(1-y)\log(1-\hat{y})\right]$$

$$= -\frac{y}{\hat{y}} - (1-y) \times \frac{1}{1-\hat{y}} \times \frac{\delta}{\delta\hat{y}}(1-\hat{y})$$

$$= -\frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}}(-1)$$

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$$= -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}$$

$$= \frac{-y + y\hat{y} + \hat{y} - y\hat{y}}{\hat{y}(1-\hat{y})}$$

$$= \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$$

$$\frac{\delta \hat{y}}{\delta z} = \frac{\delta \sigma(z)}{\delta z} = \sigma(z)(1-\sigma(z))$$

$$= \hat{y}(1-\hat{y})$$

$$\frac{\delta z}{\delta b} = \frac{\delta}{\delta b}(w_1 x_1 + w_2 x_2 + b) = 1$$

$$\frac{\delta LCE}{\delta b} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})} \times \hat{y}(1-\hat{y}) \times 1 = (\hat{y} - y)$$

$$\frac{\delta LCE}{\delta w_1} = \underbrace{\frac{\delta LCE}{\delta \hat{y}} \times \frac{\delta \hat{y}}{\delta z}}_{\text{same as before}} \times \frac{\delta z}{\delta w_1}$$

$$\frac{\delta z}{\delta w_1} = \frac{\delta}{\delta w_1}(w_1 x_1 + w_2 x_2 + b) = x_1$$

$$\frac{\delta LCE}{\delta w_1} = (\hat{y} - y)\, x_1$$

** $$\boxed{\frac{\delta LCE}{\delta w_i} = (\hat{y} - y)\, x_i}$$

$$\frac{\delta LCE}{\delta w_2} = (\hat{y} - y)\, x_2$$

Example 1

Let's assume initial weight and bias are set to zero. Learning rate is 0.1. Update the weights and bias [for one cycle] using gradient descent.

x_1	x_2	y
3	2	1

$$z = w_1 x_1 + w_2 x_2 + b$$

$$w_1 = w_2 = b = 0$$

$$w_1 = w_1 - \eta \frac{\delta L_{CE}}{\delta w_1} \rightarrow (\hat{y} - y)\, x_1$$

$$w_2 = w_2 - \eta \frac{\delta L_{CE}}{\delta w_2} \rightarrow (\hat{y} - y)\, x_2$$

$$b = b - \eta \frac{\delta L_{CE}}{\delta b} \rightarrow (\hat{y} - y)$$

$$\frac{\delta L_{CE}}{\delta w_1} = (\hat{y} - y)x_1 = (\sigma(z) - y)\, x_1 = [\sigma(w_1 x_1 + w_2 x_2 + b) - y]\, x_1$$

$$= [\sigma(0 \times 3 + 0 \times 2 + 0) - 1]\, 3$$

$$= [\sigma(0) - 1]\, 3$$

$$= (0.5 - 1)\, 3 = -1.5$$

$$\frac{\delta L_{CE}}{\delta w_2} = (\hat{y} - y)\, x_2 = (\sigma(z) - y)\, x_2 = [\sigma(0 \times 3 + 0 \times 2 + 0) - 1]\, 2$$

$$= -1$$

$$\frac{\delta L_{CE}}{\delta b}\ (\hat{y} - y) = \sigma(z) - y = \sigma(0 \times 3 + 0 \times 2 + 0) - 1 = -0.5$$

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$w_1(new) = w_1(old) - \eta(\hat{y}-y)x$

$= 0 - 0.1(-1.5) = 0.15$

$w_2(new) = w_2 - \eta(\hat{y}-y)x_2 = 0 - 0.1(-1) = 0.1$

$b = b - \eta \times (\hat{y}-y) = 0 - 0.1(-0.5) = -0.05$

Example 2

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1

$w_1 = 0.5$

$w_2 = 0.75$

$b = -1.25$

$\eta = 0.2$

$w_1 = w_1 - \eta(\hat{y}-y)x_1$

$w_2 = w_2 - \eta(\hat{y}-y)x_2$

$b = b - \eta(\hat{y}-y)$

cycle 1 (sample 1)

$x_1 = 0, \quad x_2 = 0, \quad y = 0$

$z = w_1x_1 + w_2x_2 + b = 0.5\times0 + 0.75\times0 - 1.25$

$= -1.25$

$\hat{y} = \sigma(z) = \sigma(-1.25) = 0.22 < 0.5$

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hypothesis $h_w(x) = \begin{cases} 0 & \text{if } \sigma(z) < 0.5 \\ 1 & \text{if } \sigma(z) \geq 0.5 \end{cases}$

$h_w(x) = 0 = y \Rightarrow$ currently classified

sample 2 $\Rightarrow$ $x_1 = 0,\ x_2 = 1,\ y = 1$

$$z = 0.5 \times 0 + 0.75 \times 1 - 1.25 = -0.5$$

$\hat{y} = \sigma(z) = \sigma(-0.5) = 0.38$

$h_w(x) = 0 \neq y \Rightarrow$ misclassified

↳ need to update

$w_1 = w_1 - \eta(\hat{y} - y)x_1 = 0.5 - 0.2(0.38 - 1) \times 0 = 0.5$

$w_2 = w_2 - \eta(\hat{y} - y)x_2 = 0.75 - 0.2(0.38 - 1) \times 1 = 0.87$

$b = b - \eta(\hat{y} - y) = -1.25 - 0.2(0.38 - 1) = -1.13$

sample3 $\Rightarrow$ $x_1 = 1,\ x_2 = 0,\ y = 1$

$z = w_1 x_1 + w_2 x_2 + b = 0.5 \times 1 + 0.87 \times 0 - 1.13 = \overset{-0.63}{\cancel{0.25}}$

$\hat{y} = \sigma(z) = 0.35 < 0$

$h_w(x) = 0 \Rightarrow$ misclassified

$w_1 = w_1 - \eta(\hat{y} - y)x_1 = 0.5 - 0.2(0.35 - 1) \times 1 = 0.69$

$w_2 = w_2 - \eta(\hat{y} - y)x_2 = 0.87$

$b = -1$

$z = 0.63 x_1 + 0.87 x_2 - 1$

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Sample 4: $x_1 = 1, x_2 = 1, y = 1$

$$z = 0.63 \times 1 + 0.87 \times 1 - 1.0 = 0.5$$

$$\hat{y} = \sigma(z) = \sigma(0.5) = 0.62 > 0.5$$

$h_w(x) = 1 = y \Rightarrow$ correctly classified

$w_1 = 0.63$

$w_2 = 0.87$

$b = -1.0$

MinMax Scaler (value 0-1 এর মধ্যে হবে)

$$x_i' = \frac{x_i - \min(x_i)}{\max(x_i) - \min(x_i)}$$

person	Height	Weight	Scaling	Scaled weight
A	150	50	(50-50)/(80-50)	= 0
B	160	60	(60-50)/(60-50)	= 0.33
C	170	80	(80-50)/(80-50)	= 1

min কে 0 তে convert

max কে 1 এ convert

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Standard Scaler

$$\mu_i = \frac{1}{m} \sum_{j=1}^{m} x_i^{(j)}$$

$$\sigma_i = \sqrt{\frac{1}{m} \sum_{j=1}^{m} (x_i^{(j)} - \mu_i)^2}$$

$$x_i' = \frac{x_i - \mu_i}{\sigma_i}$$

person	Height	Weight	$(x_i-\mu)^2$	Scaling	Scaled weight
A	150	$50 = x_1$	$(50-63.3)^2$	$\frac{50-63.33}{12.47}$	-1.07
B	160	$60 = x_2$	$(60-63.3)^2$	$\frac{60-63.33}{12.47}$	-0.27
C	170	$80 = x_3$	$(80-63.3)^2$	$\frac{80-63.33}{12.47}$	$+1.34$

$$\mu = \frac{50+60+80}{3} = 63.33$$

$$\sigma = \sqrt{\frac{(50-63.33)^2 + (60-63.33)^2 + (80-63.33)^2}{3}}$$

$= 12.47$

Scaled weight এর $\mu = 0$ হবে AND $\sigma = 1$ হবে ।

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Logistic Regression – Binary

Minibatch / Batch

x_1	x_2	y → binary
2	3	0
1	5	1
2	7	1

$$z = w_1x_1 + w_2x_2 + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}}$$

Given, $w_1 = 0.4$, $w_2 = 0.6$, $b = -2$. Find the predicted y

$$W = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix}$$

$$X = \begin{bmatrix} 2 & 3 \\ 1 & 5 \\ 2 & 7 \end{bmatrix} = \begin{bmatrix} x_1^1 & x_2^1 \\ x_1^2 & x_2^2 \\ x_1^3 & x_2^3 \end{bmatrix}$$

$$X.W = \begin{bmatrix} 2 & 3 \\ 1 & 5 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix} = \begin{bmatrix} 2\times0.4 + 3\times0.6 \\ 1\times0.4 + 5\times0.6 \\ 2\times0.4 + 7\times0.6 \end{bmatrix}$$

$$= \begin{bmatrix} 2.6 \\ 3.4 \\ 5.0 \end{bmatrix}$$

$$z = XW + b = \begin{bmatrix} 2.6 \\ 3.4 \\ 5.0 \end{bmatrix} + \begin{bmatrix} -2 \\ -2 \\ -2 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 1.4 \\ 3 \end{bmatrix}$$

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$$\hat{y} = \sigma(z) = \sigma\left(\begin{bmatrix} 0.6 \\ 1.4 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} \frac{1}{1+e^{-0.6}} \\ \frac{1}{1+e^{-1.4}} \\ \frac{1}{1+e^{-3}} \end{bmatrix} = \begin{bmatrix} 0.646 \\ 0.802 \\ 0.953 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \rightarrow \text{missclas}$$

Update the parameters for 1 cycle

$$w_1 = w_1 - \eta \frac{1}{3} \sum (\hat{y}_i - y_i) x_1^i \quad \rightarrow 3 \text{ datapoint}$$

$$w_2 = w_2 - \eta \frac{1}{3} \sum (\hat{y}_i - y_i) x_2^i$$

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}_{new} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}_{old} - \eta \times \frac{1}{3} \begin{bmatrix} \hat{y}_1 - y_1 \\ \hat{y}_2 - y_2 \\ \hat{y}_3 - y_3 \end{bmatrix} \cdot \begin{bmatrix} 2 & 3 \\ 1 & 5 \\ 2 & 7 \end{bmatrix}^T$$

$$= \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix} - 0.1 \times \frac{1}{3} \begin{bmatrix} 2 & 1 & 2 \\ 3 & 5 & 7 \end{bmatrix} \cdot \begin{bmatrix} 0.646 - 0 \\ 0.802 - 1 \\ 0.953 - 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix} - \frac{0.1}{3} \begin{bmatrix} 2 \times 0.646 - 1 \times 0.198 - 2 \times 0.0.047 \\ 3 \times 0.646 - 5 \times 0.198 - 7 \times 0.047 \end{bmatrix}$$

$$= \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix} - \frac{0.1}{3} \begin{bmatrix} 1 \\ 0.619 \end{bmatrix} = \begin{bmatrix} 0.367 \\ 0.579 \end{bmatrix}$$

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$$b = b - \eta \frac{1}{3} \Sigma \left(\hat{y}_i - y_i\right)$$

need to update b also.

Logistic Regression – Multiclass

x_1	x_2	y
2	1	0
3	4	1
1	5	2

number of class, $k = 3$

number of datapoint, $m = 3$

feature, $f = 2$

x_1	x_2	y	y_1 ($\nearrow z_1$)	y_2 ($\nearrow z_2$)	y_3 ($\nearrow z_3$)
2	1	0	1	0	0
3	4	1	0	1	0
1	5	2	0	0	1

$$\begin{bmatrix} z_1^1 & z_2^1 & z_3^1 \\ z_1^2 & z_2^2 & z_3^2 \\ z_1^3 & z_2^3 & z_3^3 \end{bmatrix}$$

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x_1	x_2	y	y_1	y_2	y_3
2	1	0	1	0	0

$z_1 \quad z_2 \quad z_3$

$$\begin{bmatrix} z_1 & z_2 & z_3 \end{bmatrix} \xrightarrow{\text{softmax}} \begin{bmatrix} \hat{y}_1 & \hat{y}_2 & \hat{y}_3 \end{bmatrix}$$

$$z_1 = W_{11} x_1 + W_{12} x_2 + b_1$$

$$z_2 = W_{21} x_1 + W_{22} x_2 + b_2$$

$$z_3 = W_{32} x_1 + W_{32} x_2 + b_3$$

$$\hat{y}_1 = \text{softmax}(z_1) = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

$$\hat{y}_2 = \text{softmax}(z_2) = \frac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

$$\hat{y}_3 = \text{softmax}(z_3) = \frac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

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x_1	x_2	y
2	1	0
3	4	1
1	5	2

$b = [0.1 \quad -0.2 \quad 0]$

0 → class y_1

1 → y_2

2 → y_3

→ Given

$$W = \begin{array}{c} \begin{array}{ccc} z_1 & z_2 & z_3 \end{array} \\ \begin{bmatrix} w_{11} & w_{21} & w_{31} \\ w_{12} & w_{22} & w_{32} \end{bmatrix} \end{array} = \begin{bmatrix} 0.2 & -0.1 & 0.3 \\ 0.4 & 0.5 & -0.2 \end{bmatrix}$$

$$X = \begin{bmatrix} x_1^1 & x_2^1 \\ x_1^2 & x_2^2 \\ x_1^3 & x_2^3 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 4 \\ 1 & 5 \end{bmatrix}$$

$$X \cdot W = \begin{bmatrix} 2 & 1 \\ 3 & 4 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 0.2 & -0.1 & 0.3 \\ 0.4 & 0.5 & -0.2 \end{bmatrix}$$

$$= \begin{bmatrix} 0.8 & 0.3 & 0.4 \\ 2.2 & 1.7 & 0.1 \\ 2.2 & 2.4 & -0.7 \end{bmatrix}$$

$$XW + b = \begin{bmatrix} 0.8 & 0.3 & 0.4 \\ 2.2 & 1.7 & 0.1 \\ 2.2 & 2.4 & -0.7 \end{bmatrix} + \begin{bmatrix} 0.1 & -0.2 & 0 \\ 0.1 & -0.2 & 0 \\ 0.1 & -0.2 & 0 \end{bmatrix}$$

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$$= \begin{bmatrix} 0.9\, z_1^{1} & 0.1\, z_2^{1} & 0.4\, z_3^{1} \\ 2.3\, z_1^{2} & 1.5\, z_2^{2} & 0.1\, z_3^{2} \\ 2.3\, z_1^{3} & 2.1\, z_2^{3} & -0.2\, z_3^{3} \end{bmatrix}$$

Then softmax

$$\hat{y} = \text{softmax}(z) = \begin{bmatrix} \frac{e^{z_1^1}}{e^{z_1^1}+e^{z_2^1}+e^{z_3^1}} & \frac{e^{z_2^1}}{e^{z_1^1}+e^{z_2^1}+e^{z_3^1}} & \frac{e^{z_3^1}}{e^{z_1^1}+e^{z_2^1}+e^{z_3^1}} \\ \frac{e^{z_1^2}}{e^{z_1^2}+e^{z_2^2}+e^{z_3^2}} & \frac{e^{z_2^2}}{e^{z_1^2}+e^{z_2^2}+e^{z_3^2}} & \frac{e^{z_3^2}}{e^{z_1^2}+e^{z_2^2}+e^{z_3^2}} \\ \frac{e^{z_1^3}}{e^{z_1^3}+e^{z_2^3}+e^{z_3^3}} & \frac{e^{z_2^3}}{e^{z_1^3}+e^{z_2^3}+e^{z_3^3}} & \frac{e^{z_3^3}}{e^{z_1^3}+e^{z_2^3}+e^{z_3^3}} \end{bmatrix}$$

$$= \begin{bmatrix} 0.4866\, \hat{y}_1^{1} & 0.2186\, \hat{y}_2^{1} & 0.2948\, \hat{y}_3^{1} \\ 0.6412\, \hat{y}_1^{2} & 0.2881\, \hat{y}_2^{2} & 0.0709\, \hat{y}_3^{2} \\ 0.5117\, \hat{y}_1^{3} & 0.4630\, \hat{y}_2^{3} & 0.0255\, \hat{y}_3^{3} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \text{class } y_1 \\ \text{class } y_1 \\ \text{class } y_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$